



Techno Bengal Institute of Technology

(Formerly known as Bengal Institute of Technology)

College Code:121



TOPIC: DC TRANSIENT AND RL AND RC CIRCUITS

CA1 Examination, Odd Semester, 2025-26

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Semester	1ST
Paper Code	ES - EE101
Paper Name	BASIC ELECTRICAL ENGINEERING

DC TRANSIENT AND RL & RC CIRCUITS

DC transients refer to the temporary change in current and voltage that occurs in RC and RL circuits when a DC supply is switched ON or OFF. In RC circuits, the transient response is governed by the charging and discharging of the capacitor, while in RL circuits it is governed by the build-up and decay of current through the inductor due to energy stored in its magnetic field.

DC TRANSIENT

- A DC transient is a temporary change in current or voltage that occurs when a DC circuit is switched ON or OFF.
- It exists only for a short duration before the circuit reaches steady state.
- Transients occur due to the presence of energy storage elements like inductors and capacitors.
- Inductors oppose sudden change in current, while capacitors oppose sudden change in voltage.
- The duration of the transient depends on the time constant of the circuit.
- DC transients are commonly observed in RL and RC circuits.

EXAMPLES OF DC TRANSIENT

Switching ON/OFF an RL circuit :

Current changes gradually due to the inductor opposing sudden change.

Charging of a capacitor in an RC circuit :

Capacitor voltage increases gradually after DC supply is applied.

Discharging of a capacitor :

Stored energy is released through a resistor, causing a transient current.

Relay or solenoid operation :

Sudden switching produces transient currents and induced voltages.

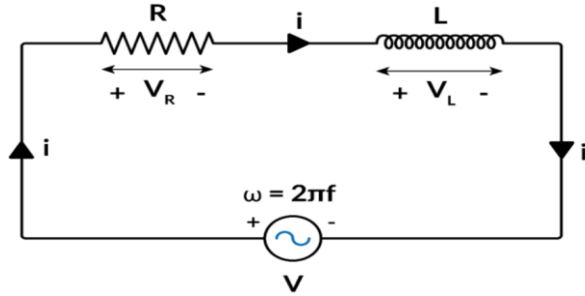
DC motor starting :

High initial current and gradual stabilization due to inductance.

ANALYSIS ON RL CIRCUIT

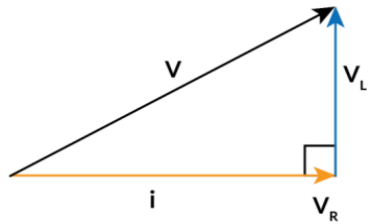
Series RL Circuit

The circuit diagram for the series RL circuit is as follows :

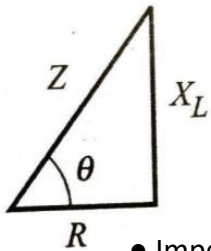


The important takeaways to be remembered are as follows :

- Voltage equation: $v = v_R + v_L$
- Current equation: $i = i_R = i_L$
- Phasor diagram: The following diagram represents the phasor diagram of series RL circuits.



- Impedance triangle: The following figure represents the impedance triangle.



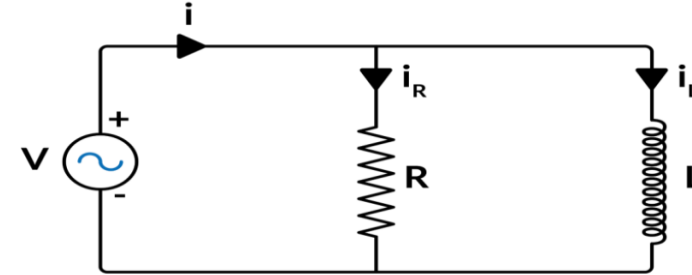
◦ Impedance: $Z = R + jX_L$

◦ Magnitude of impedance: $|Z| = \sqrt{R^2 + X_L^2}$

- Impedance angle: $\theta = \tan^{-1}(X_L/R)$
- Power factor: $\cos \theta = R/Z$ lagging .

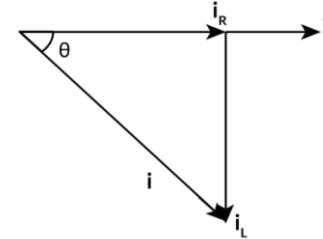
Parallel RL Circuit

The circuit diagram for the parallel RL circuit is given as follows.

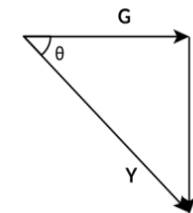


The important takeaways to be remembered are as follows:

- Current equation: $i = i_R + i_L$
- Voltage equation: $v = v_R = v_L$
- Phasor diagram: The phasor diagram for parallel RL circuits is given as follows.



- Admittance triangle: The admittance triangle for series RL circuits is given as follows.



◦ Admittance: $Y = G - jB_L$

◦ Magnitude of admittance: $|Y| = \sqrt{G^2 + B_L^2}$

Where,

Y = admittance = $1/Z$

G = conductance = $1/R$

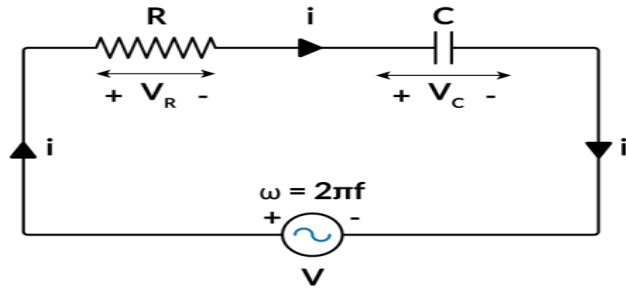
B_L = Inductive susceptance = $1/X_L$

- Admittance angle: $\theta = \tan^{-1}(B_L/G)$
- Power factor: $\cos \theta = G/Y$ lagging

Analysis of RC Circuits

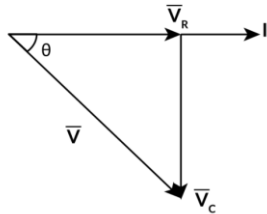
Series RC Circuit

The circuit diagram for series RC circuits is given as follows

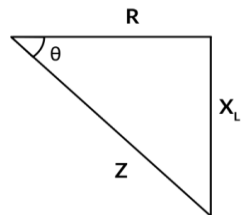


The important takeaways to be remembered are as follows:

- Current equation: $i = i_R = i_C$
- Voltage equation: $v = v_R + v_C$
- Phasor diagram: The phasor diagram for the series RC circuit is given as follows.



- Impedance triangle: The impedance triangle of series RC circuit is given as follows.

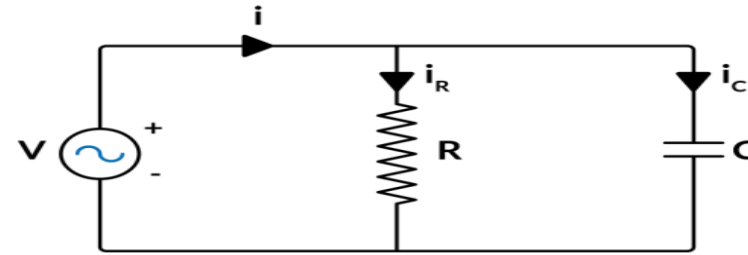


- Impedance: $Z = R - jX_C$
- Magnitude of impedance: $|Z| = \sqrt{R^2 + X_C^2}$

- Impedance angle: $\theta = \tan^{-1}(X_C/R)$
- Power factor: $\cos \theta = R/Z$ leading

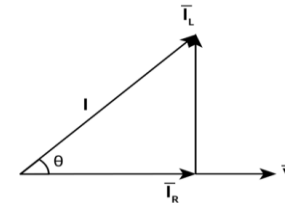
Parallel RC Circuit

The circuit diagram for the parallel RC circuit is given as follows.

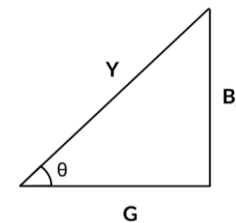


The important takeaways to be remembered are as follows:

- Voltage equation: $v = v_R = v_C$
- Current equation: $i = i_R + i_C$
- Phasor diagram: The phasor diagram of parallel RC circuits is given as follows.



- Admittance triangle: The admittance triangle for series RL circuit is given as follows.



- Admittance: $Y = G + jB_C$
- Magnitude of admittance: $|Y| = \sqrt{G^2 + B_C^2}$
- Where $B_C = \text{capacitive susceptance} = 1/X_C$

- Admittance angle: $\theta = \tan^{-1} B_C/G$
- Power factor: $\cos \theta = G/Y$ leading

RL VS RC CIRCUIT DIFFERENCE

ASPECT	RC CIRCUIT	RL CIRCUIT
Main Components	Resistor (R) and Capacitor (C)	Resistor (R) and Inductor (L)
Energy Storage Element	Capacitor stores energy in an electric field	Inductor stores energy in a magnetic field
Opposition to Change	Capacitor opposes sudden change in voltage	Inductor opposes sudden change in current
Current at Switching ON (DC)	Maximum initially, then decreases to zero	Zero initially, then increases to steady value
Voltage at Switching ON (DC)	Starts from zero, rises gradually	Maximum initially, then decreases
Time Constant	$\tau = RC$	$\tau = L / R$
Behavior in DC Steady State	Capacitor behaves like an open circuit	Inductor behaves like a short circuit
Phase Relation (AC)	Current leads voltage	Current lags voltage
Typical Applications	Filters, timing circuits, coupling circuits	Motors, chokes, power supplies

CONCLUSION

DC transient analysis of RC and RL circuits explains how current and voltage change with time immediately after a DC source is switched ON or OFF. RC circuits emphasize voltage variation due to capacitor charging and discharging, while RL circuits emphasize current variation due to the magnetic energy stored in the inductor. Understanding these transient behaviors is essential for the proper design and analysis of electrical and electronic systems such as filters, timing circuits, and power control applications.

Thank You